

1. Introduction

This work performs the initial steps that are required to create a predictive customer churn machine learning model. The goals of this initial analysis is to combine multiple data sources relevant to the customer behavior, perform exploratory data analysis(EDA) to uncover some patterns associated with customer churn, clean and preprocess the dataset and lastly is to engineer some features to improve the accuracy of the machine learning models

2. Data Sources and Rationale

The included datasets in the Excel file given are:

- **Customer_Demographics** - this consists of personal attributes for each customers
- **Transaction_History** - this consists of dates and amounts of transactions which captures spending behavior, recency and tenure of each customer.
- **Customer_Service** - this consists of records of customer support interactions' dates and types. This can be used as a signal leading to churn or engagements preventing churn.
- **Online_Activity** - this consists of the login frequency, last login and service usage of the customers. This can help for identifying and predicting customer retention.
- **Churn_Status** - this is a binary label representing whether each customer churned.

Overall, these datasets provides a view of customer activity across purchasing, customer support and online engagement which can be the core determinants of customer churn.

3. Data Processing Summary

After loading the data, it was read and merged on CustomerID. The merge used was a left-join onto the demographics to ensure the attributes will align per customer. Aggregate tables were created from transactional and service logs (e.g., first & last transaction dates, total spend, transaction counts, total service interactions, service diversity).

Next thing I did was to convert the dates to datetime and a consistent reference table which is the max TransactionDate within the dataset was used to compute the recency metrics, ensuring that the recency is connected to the available data

After that, I added and engineered features such as:

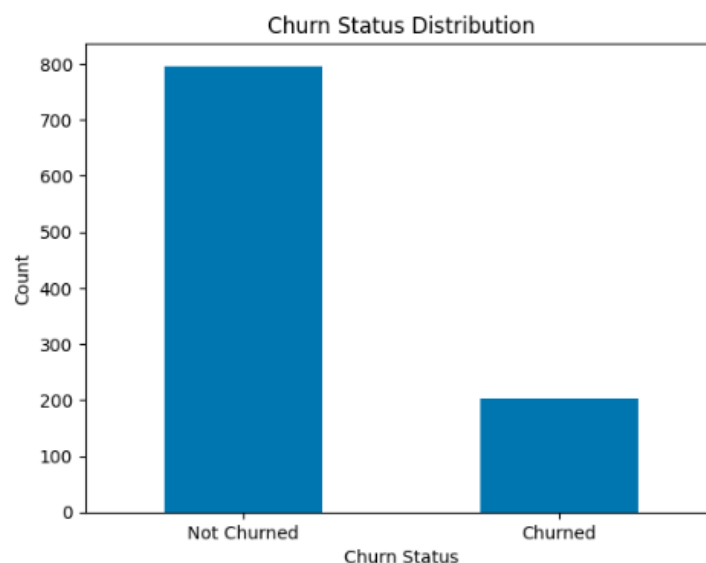
- **FirstTransactionDate** - this establishes the start of the customer relationship that is useful for measuring lifetime value and tenure.
- **LastTransactionDate** - this captures the most recent engagement which can be a strong indicator of whether a customer is still active.
- **TotalAmountSpent** - this measures total customer value which helps identify high-spending churn risks

- AvgAmountPerTransaction - this distinguishes between frequent small spenders and infrequent big spenders.
- TransactionCount - this indicates the engagement frequency like how fewer transactions can signal a churn risk.
- TotalServiceInteractions - high counts may indicate dissatisfaction or heavy engagement; context matters.
- ServiceDiversity - this shows the variety of service interactions a customer did.
- RecencyDays - this shows the days since last purchase; one of the strongest early-warning churn signals.
- TenureDays - this may show how long the customer has been using the service for example, longer tenure may mean loyalty, but churn can still occur after long relationships.
- DaysSinceLastLogin - this tracks online disengagement; rising numbers often precede churn.

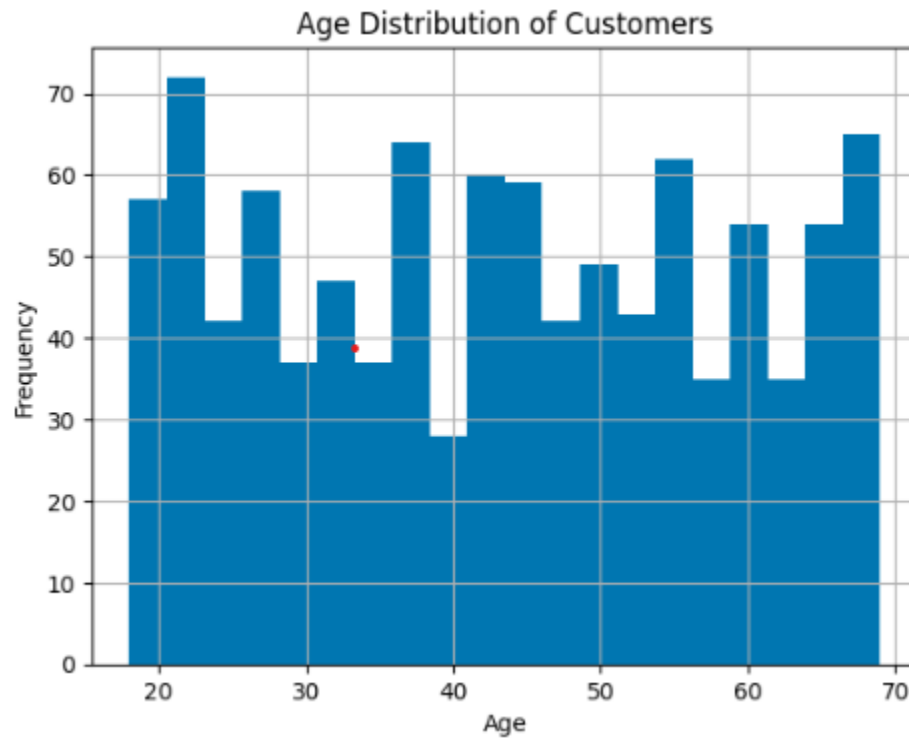
For the missing values, count columns like the TotalServiceInteractions were set to 0 when missing, while interaction dates like LastInteractionDate will be using proxy such as LastLoginDate if their value is missing and likewise towards FirstInteractionDate and FirstTransactionNDate. Then for the numeric columns, missing data were filled with medians before they were scaled.

After that, TotalAmountSpent and TransactionCount were winsorized and capped at the 99th percentile to reduce skew caused by extreme spenders while preserving relative order for the model. With the initial cleaning done, categorical variables like MaritalStatus, Gender, IncomeLevel and ServiceUsage were one-hot encoded. Numeric features were standardized using StandardScaler for machine learning algorithms sensitive to scale. The scaler was saved for consistent preprocessing during model training and inference

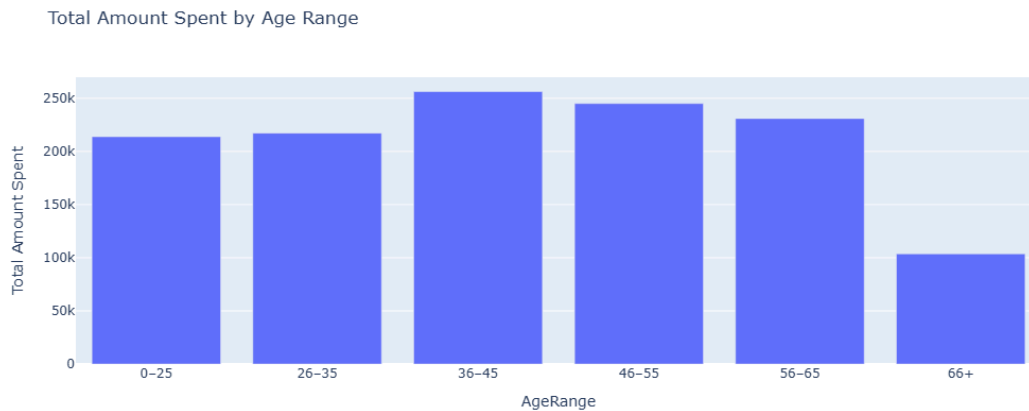
4. EDA Highlights



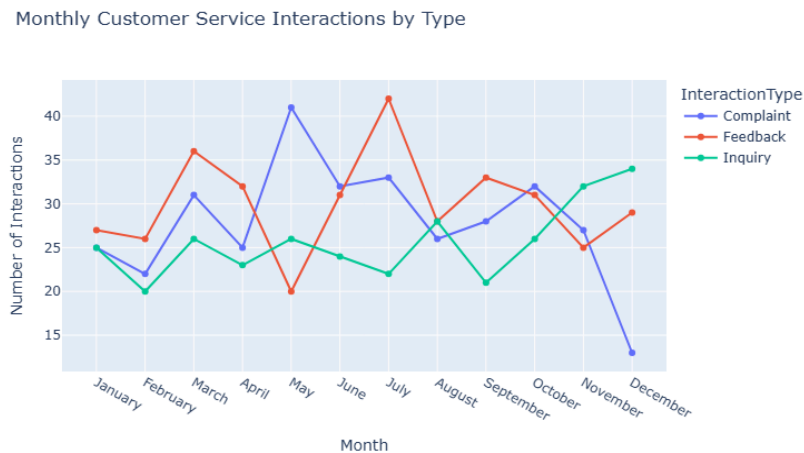
The dataset shows a balanced churn distribution, with a substantial portion of customers remaining loyal while a significant percentage churned. This provides a clear starting point for investigating the elements that influence client retention and attrition. Understanding the behavioral and demographic variations between these two groups is critical for creating successful retention strategies.



The age distribution of customers reveals that most of them are in the young to middle-aged age range, especially those between the ages of 25 and 44. In addition to indicating that marketing and service offerings may already be appealing to this age group, this demographic concentration emphasizes the need of looking into ways to draw in both younger and older clients in order to attain a more balanced age representation.

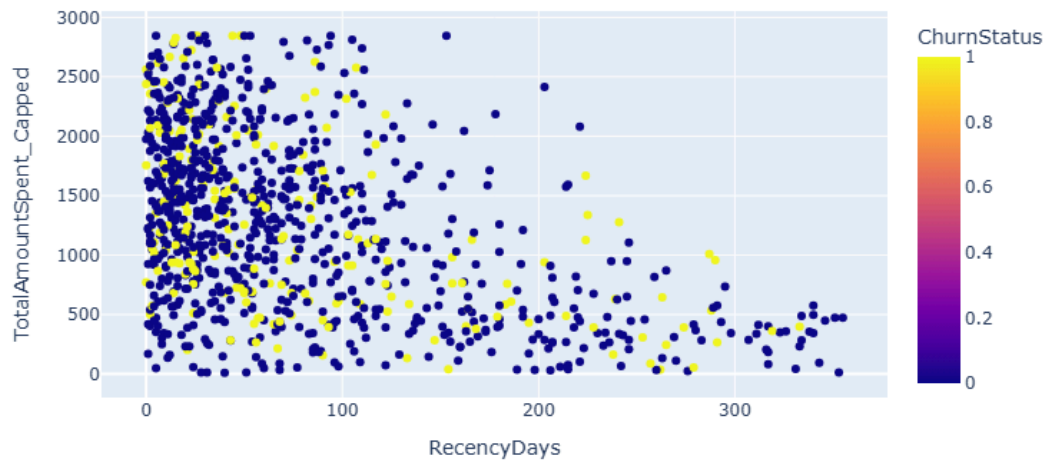


When spending behavior is examined by age group, the data reveals significant disparities in mean expenditure amounts. Certain age groups, particularly those aged 35 to 54, tend to spend more on average, implying a greater potential for revenue production from this segment. This information can be used to steer customized promotions, loyalty programs, and personalized offers that maximize value for high-spending groups while nurturing lower-spending segments.



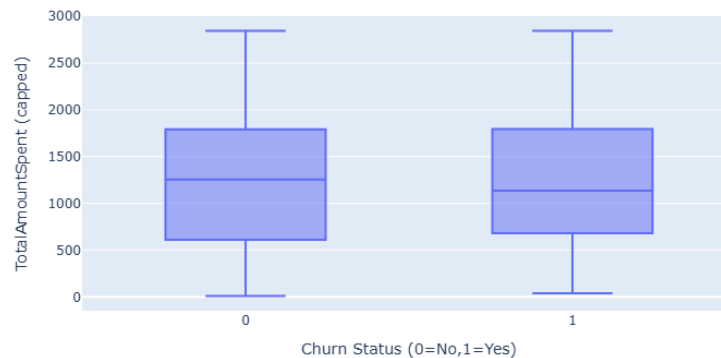
Analysis of client interactions by type shows that some channels are utilized more frequently than others, which may represent customer preferences or the effectiveness of specific connections. For example, if digital channels or specific service types are dominant, organizations might prioritize these channels while enhancing underutilized ones to increase overall involvement.

Recency vs Total Amount Spent (sampled)



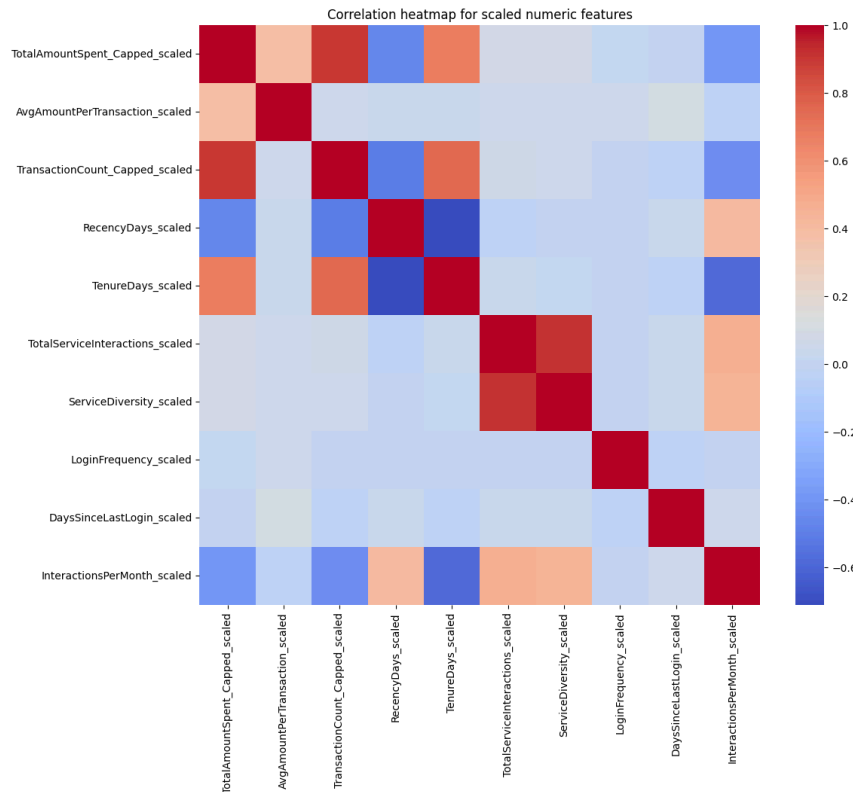
The scatter plot of RecencyDays against TotalAmountSpent_Capped reveals a unique trend in which larger spending is concentrated among consumers with low recency values, indicating that they purchased lately. As recency increases (customers haven't purchased in a long time), the overall cost tends to fall, with fewer high-spending outliers. The color mapping for churn status shows that churned customers (yellow dots) appear throughout the recency range, although they are more noticeable among those with higher recency and lower expenditure. This trend implies that recent purchase activity is a good predictor of engagement, and higher recency mixed with low spending may indicate increased churn risk.

Total Amount Spent (Capped) by Churn Status



The box plot comparing TotalAmountSpent_Capped by churn status indicates that non-churned customers (status 0) have higher median spending than churned customers (status 1). Non-churned clients have a somewhat wider interquartile range, indicating that their spending varies more. However, both groups exhibit identical

maximum expenditure numbers, implying that even churned customers may have had significant prior spending. The primary distinction is that churned customers tend to spend less, which supports the notion that reduced total spending can be a warning indicator of churn.



The correlation heatmap shows how the dataset's scaled numeric features relate to each other. A major positive connection exists between `TotalAmountSpent_Capped_scaled` and `TransactionCount_Capped_scaled`, showing that consumers who conduct more transactions spend more overall. Similarly, `TenureDays_scaled` shows a significant positive association with `TransactionCount_Capped_scaled`, implying that longer-tenured customers have larger transaction counts. `RecencyDays_scaled`, on the other hand, has a negative association with both `TotalAmountSpent_Capped_scaled` and `TransactionCount_Capped_scaled`, implying that customers who purchased recently spend more and engage in more transactions. Most other variables have modest correlations, indicating that they give unique, less redundant information.

5. Files produced

- **customer_churn_analysis_cleaned.csv** — cleaned, feature-engineered, encoded, and partially scaled dataset prepared for model-building.
- **numeric_scaler.save** — `StandardScaler` object to apply the same scaling during model training and production inference.

6. Conclusion

The cleaned and feature-engineered dataset is now ready for model training. The preprocessing decisions were selected to strike a balance between business sense and statistical robustness. The next step is to create classification models, tune hyperparameters, and run explainability tests (SHAP) to identify actionable retention methods.